YESHIVA COLLEGE COMPUTER SCIENCE

The B.S. in Computer Science

Tracks, course schedules and student projects: the details behind our open-house poster. Updated October 2026.

The major at a glance

- **Degree:** Bachelor of Science (B.S.) in Computer Science.
- **Two tracks:** Distributed Systems (20 courses, 67 credits) and Artificial Intelligence (22 courses, 75 credits).
- **Four years:** an 8-semester program; 128 total credits to graduate, including 8 Yeshiva College Core courses.
- **Senior capstone:** every student builds a complete, real-world system, and may combine it with a Yeshiva College honors thesis.
- **Industry-designed:** the curriculum was developed with an industry advisory board of leaders from Google, Amazon, Goldman Sachs, IBM, Two Sigma, Mayo Clinic and more (listed below).

Choosing a track

	Distributed Systems	Artificial Intelligence
General software engineering	Best	Better
Cloud computing	Best	Good
Cybersecurity	Best	—
Machine learning	—	Best
Artificial intelligence, natural language processing	Good	Best
Courses / credits	20 / 67	22 / 75
I should choose this track if…	I enjoy building	I enjoy analyzing
Is graduate school necessary to get, and keep, great jobs?	No	No

"Building vs. analyzing" is a guide to inclinations, not a strict divide: AI students build, and Distributed Systems students analyze.

Distributed Systems track

Learn to build the large-scale software systems that almost every major industry depends on: internet companies (Google, Amazon, Netflix), finance (trading, high-speed risk modeling), logistics, data-driven medicine (Mayo Clinic, Memorial Sloan Kettering) and transportation. The track covers:

- Computer science and software engineering fundamentals
- In-depth, applied knowledge of every level of the software stack: system, operating system, network, application, database and distributed
- Building software at any scale, from a single process to systems distributed across the internet (cloud computing)
- The architecture and proper use of SQL, NoSQL and NewSQL databases
- Building secure software

Artificial Intelligence track

Data-rich industries, such as tech, finance, marketing and logistics, increasingly use statistical and probabilistic approaches in mission-critical decisions and software systems. The track covers:

- Computer science and software engineering fundamentals
- In-depth, applied knowledge of artificial intelligence, machine learning, computer vision and natural language processing
- The mathematics behind AI: calculus, linear algebra, probability and statistics
- Taking machine learning from prototype to production (ML engineering)

Semester-by-semester schedules

Distributed Systems track (20 courses, 67 credits)

Year	Fall semester	Spring semester
1st	Intro to C.S. (COM 1300) Calculus I (MAT 1412) YC Core #1	Data Structures (COM 1320) Linear Algebra (MAT 2105) Mathematics for Computer Science (COM 1310) YC Core #2
2nd	Intro to Algorithms (COM 2545) Computer Organization (COM 2113) YC Core #3	Design & Analysis of Algorithms (COM 2546) Operating Systems (COM 3610) YC Core #4
3rd	Distributed Systems (COM 3800) Parallel Programming (COM 3820) Networking (COM 2512) YC Core #5	Advanced Distributed Systems (COM 3810) Cybersecurity (COM 4580) Modern Data Management (COM 3580) YC Core #6
4th	Programming Languages (COM 3640) Database Implementation (COM 3563) Artificial Intelligence (COM 3760) YC Core #7 (elective)	Compilers & Tools (COM 3645) Capstone Project (COM 4020) YC Core #8 (elective)

Artificial Intelligence track (22 courses, 75 credits)

Year	Fall semester	Spring semester
1st	Intro to C.S. (COM 1300) Calculus I (MAT 1412) YC Core #1	Data Structures (COM 1320) Calculus II (MAT 1413) Mathematics for Computer Science (COM 1310) YC Core #2
2nd	Introduction to Algorithms (COM 2545) Linear Algebra (MAT 2105) Computer Organization (COM 2113) YC Core #3	Design & Analysis of Algorithms (COM 2546) Multivariable Calculus (MAT 1510) Probability Theory (MAT 2461) YC Core #4
3rd	Artificial Intelligence (COM 3760) Mathematical Statistics (MAT 2462) Programming Languages (COM 3640) YC Core #5	Machine Learning (COM 3920) Modern Data Management (COM 3580) Operating Systems (COM 3610) YC Core #6
4th	Distributed Systems (COM 3800) Advanced Machine Learning (COM 4010) Parallel Algorithms & Programming (COM 3820) YC Core #7 (elective)	Natural Language Processing (COM 3930) Capstone Project (COM 4020) YC Core #8 (elective)

Students without 32 credits from a YU Israel program take additional courses to reach the 128 credits required to graduate, and plan their schedule with an academic advisor.

Advanced courses

Distributed Systems (COM 3800): How networked computers combine into highly available, scalable services: time synchronization, coordination and consensus, replication, fault tolerance, cluster scheduling and cloud computing architecture.

Advanced Distributed Systems (COM 3810): Implementing and operating scalable, fault-tolerant systems: instrumentation and monitoring, performance and SLAs, and detecting and remedying failures, using widely used architectures.

Parallel Algorithms & Programming (COM 3820): Concurrent programming and parallel algorithm design: locality, shared vs. non-shared memory, synchronization, and programming models from threads to message passing.

Cybersecurity (COM 4580): Cryptography, authentication and access control, malware, web vulnerabilities, network security, and designing and coding secure software, on both the offensive and defensive sides.

Networking (COM 2512): How the internet and the services on it work and why: protocols, transport, routing, wireless and mobile networks, and network security; building reliable networked applications.

Modern Data Management (COM 3580): The relational model and SQL, NoSQL stores, streaming and messaging systems, Apache Spark, and building data pipelines across multiple kinds of data systems.

Database Implementation (COM 3563): How relational databases work inside: storage, buffer management, indexing, query planning and execution, transactions, recovery and concurrency control.

Programming Languages (COM 3640) and Compilers & Tools (COM 3645): Language design, syntax and semantics across paradigms; how code becomes machine instructions (parsing, semantic analysis, code generation); build tools and domain-specific languages.

Artificial Intelligence (COM 3760): Search, constraint satisfaction, game playing, knowledge representation, planning, and reasoning and decision-making under uncertainty.

Machine Learning (COM 3920): Logistic regression, neural networks and CNNs, decision trees and ensembles, NLP, reinforcement and unsupervised learning, with the open-source libraries and cloud ML services practitioners use.

Natural Language Processing (COM 3930): Computational methods for extracting meaning from text: n-grams, text mining, search engines and information retrieval, and word embeddings.

Advanced Machine Learning (COM 4010): Machine-learning engineering: the ML lifecycle, deploying models to production, and big data in real-world ML applications.

Capstone Project (COM 4020): Under faculty supervision, students build a complete, real-world system end to end, following a full software-engineering lifecycle. Students may combine the capstone with a Yeshiva College honors thesis.

Industry advisory board

Our curriculum was developed in consultation with these industry leaders:

- **Dr. Henrique Andrade**, Principal Engineer, Shopify
- **Dr. Erick Brethenoux**, Chief of Research, Artificial Intelligence, Gartner; adjunct professor, Stuart School of Business, Illinois Institute of Technology

- **Steve Demuth**, CTO, Mayo Clinic (retired)
- **Rich Dutton**, Senior Staff Software Engineer, Cloud Machine Learning Scaling, Google
- **Dr. Aliza Heching**, Research Staff Member, IBM Research
- **Dr. Jonathan Hosking**, Principal Research Scientist, Amazon
- **Paul McGregor**, Technology Fellow and Managing Director, Goldman Sachs
- **Dr. Howard Morgan**, Chairman, B Capital Group
- **Dr. Claudia Perlich**, Head of Strategic Data Science, Investment Management, Two Sigma
- **Dr. Kavitha Srinivas**, Research Staff Member, IBM
- **Dr. Joel Wein**, Senior Director of Engineering, Infrastructure Storage, Google

Student work at the FINOS Open Source in Finance Forum

November 5, 2026 · Open Source in Finance Forum (OSFF), New York · presented by FINOS and the Linux Foundation

Prof. Judah Diament and students Yaakov Seif and Alex Neugroschl presented “Fulfilling the TraderX Vision,” showing how undergraduates used spec-driven development and LLM coding agents to grow FINOS TraderX, an open-source reference trading system, into a consensus-replicated, multi-asset trading venue, and plugged in a separately developed market-risk engine. One student started in June with no background in finance.

The results

- 18 new spec-driven TraderX versions (YU01–YU18), built June 9 to October 22, 2026
- A matching engine that handles 2.41 million orders per second (in process, one thread)
- More than 300,000 orders per second, one order at a time over FIX and binary protocols, through a three-member Raft consensus cluster on Google Kubernetes Engine
- A new leader elected in about 0.63 seconds under load, with no acknowledged order lost
- 961 Java unit test cases passing
- A market-risk engine in JAX, ported from the Open Source Risk Engine (ORE) and validated against it, pricing swaps, swaptions and Treasuries and computing exposure, VaR, expected shortfall and Greeks, run against TraderX positions on Google Cloud

Supported by Google

The Computer Science Department received a **grant from Google** to support this research, including work to make risk calculations faster and cheaper using JAX and Google TPUs.

What comes next

- Fall 2026–Spring 2027: faster, cheaper risk calculations using JAX and Google TPUs
- Spring 2027: senior capstone projects focused on financial systems
- Beyond: large-scale financial systems work built into courses such as Advanced Distributed Systems, AI/ML, Programming Languages, Compilers, Modern Data Management and Cybersecurity

Learn more: traderx.finos.org · student code: github.com/YU-CS-Summer-2026-Distributed-TraderX/traderX

AI research with Memorial Sloan Kettering Cancer Center

Our students work with researchers at Memorial Sloan Kettering Cancer Center (MSKCC) on two projects:

- **Computer vision:** building models that detect and track tumors.
- **Large language models for automated quality assurance of oncology radiology reports:** students deployed LLMs and built interfaces to run inference, worked with large-scale data distribution, processing and splits for model training and validation, engineered prompts for the project's use cases, and set up performance metrics and implemented them on MSKCC's annotation platform for model validation.

Students working on these projects use GPU servers housed at Yeshiva University as well as MSKCC's compute cluster.

Learn more

- Department website: yu.edu/yc/cs
- Frequently asked questions: yu.edu/yeshiva-college/ug/computer-science
- The complete Guide to the Computer Science Major is available from the department.